

# Set Theory Enderton - Solutions Manual

Mingruifu Lin

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# Chapter 1

## Introduction

### Exercise 1

- (a) Both  $\in$  and  $\subseteq$ .
- (b) Only  $\subseteq$ .
- (c) Only  $\subseteq$ .
- (d) Only  $\in$ .
- (e) None works.

### Exercise 2

The empty set contains no element, whereas the others both contain something. The second set contains the empty set, but the third does not. Hence, they are pairwise unequal.

### Exercise 3

$$a \in \mathcal{P}(B) \Rightarrow a \subseteq B \Rightarrow a \subseteq C \Rightarrow a \in \mathcal{P}(C)$$

where the second implication comes from the transitivity of the subset relation.

### Exercise 4

$$x, y \in B \Rightarrow \{x, y\}, \{x\} \in \mathcal{P}(B) \Rightarrow \{\{x, y\}, \{x\}\} \in \mathcal{PP}(B)$$

### Exercise 5

The rank of  $\{\{\emptyset\}\}$  is 2. The rank of  $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$  is 3.

### Exercise 6

I don't understand the question.

**Exercise 7**

Fuh naw.

## Chapter 2

# Axioms and Operations

### Exercise 1

Set of integers divisible by 180.

### Exercise 2

Let  $A = \{\{1\}, \emptyset\}$  and  $B = \{\{1\}\}$ .

### Exercise 3